

- 1 (a) Explain what is meant by the accuracy of a measured value.

.....
 [1]

- (b) Two solid cubes, A and B, are measured to determine the density of their materials.

Table 1.1 shows the measurements for cube A.

Table 1.1

quantity	measurement
length of side	$(1.53 \pm 0.01) \text{ cm}$
mass	$(31.3 \pm 0.5) \text{ g}$

- (i) Show that the calculated density of the material of cube A is $8.7 \times 10^3 \text{ kg m}^{-3}$.

[2]

- (ii) Calculate the percentage uncertainty in the density of the material of cube A.

percentage uncertainty =% [2]

- (iii) The density of the material of cube B is determined to be $9.2 \times 10^3 \text{ kg m}^{-3} \pm 6\%$.

State and explain whether cube A and cube B could be made from the same material.

.....

 [2]

[Total: 7]



- 1 Scientists are investigating the variation in air pressure at different locations on a mountain.

- (a) The scientists take measurements of several physical quantities at each location.

Complete Table 1.1 by stating the SI base unit for each quantity and identifying with a tick (✓) whether each quantity is a scalar or a vector. Use the space for any working.

Table 1.1

quantity measured	SI base unit	scalar	vector
air temperature			
air pressure			

[2]

- (b) (i) At one location, the density of the air is 1.1 kg m^{-3} . A spherical weather balloon is filled with a gas and released from rest. The balloon has radius 0.90 m.

Calculate the upthrust acting on the balloon when it is released.

upthrust = N [2]

- (ii) Explain why an upthrust acts on the balloon.

.....

 [2]



(iii) The balloon has weight 19 N.

Calculate the magnitude of the initial acceleration of the balloon.

acceleration = ms^{-2} [3]

- (c) A quantity c relating to the motion of the balloon is calculated from three measured quantities k , F and v using the formula

$$c = \frac{2kF}{v^2}.$$

The percentage uncertainties in the measured quantities are given in Table 1.2.

Table 1.2

measured quantity	percentage uncertainty
k	5%
F	3%
v	4%

The calculated value of c is 1.8.

Determine the absolute uncertainty in c .

absolute uncertainty = [2]

[Total: 11]

